# A Technical Introduction to Cross-validation statistics

## Section 1

Measures of fit The goal of cross-validation is to estimate the expected level of fit of a model to a data set that is independent of the data that were used to train the model. It can be used to estimate any quantitative measure of fit that is appropriate for the data and model. For example, for binary classification problems, each case in the validation set is either predicted correctly or incorrectly. In this situation the misclassification error rate can be used to summarize the fit, although other measures derived from information (e.g., counts, frequency) contained within a contingency table or confusion matrix could also be used. When the value being predicted is continuously distributed, the mean squared error, root mean squared error or median absolute deviation could be used to summarize the errors.

## Section 2

Leave-one-out cross-validation (LOOCV) is a particular case of leave-p-out cross-validation with p = 1. The process looks similar to jackknife; however, with cross-validation one computes a statistic on the left-out sample(s), while with jackknifing one computes a statistic from the kept samples only. LOO cross-validation requires less computation time than LpO cross-validation because there are only C 1 n = n {\displaystyle C_{1}^{n}=n} passes rather than C p n {\displaystyle C_{p}^{n}} . However, n {\displaystyle n} passes may still require quite a large computation time, in which case other approaches such as k-fold cross validation may be more appropriate. Pseudo-code algorithm: Input: x, {vector of length N with x-values of incoming points} y, {vector of length N with y-values of the expected result} interpolate( x_in, y_in, x_out ), { returns the estimation for point x_out after the model is trained with x_in-y_in pairs} Output: err, {estimate for the prediction error} Steps:

## Section 3

Nested cross-validation When cross-validation is used simultaneously for selection of the best set of hyperparameters and for error estimation (and assessment of generalization capacity), a nested cross-validation is required. Many variants exist. At least two variants can be distinguished:

## Section 4

Hoornweg (2018) shows that a loss function with such an accuracy-simplicity tradeoff can also be used to intuitively define shrinkage estimators like the (adaptive) lasso and Bayesian / ridge regression. Click on the lasso for an example.

## Section 5

k-fold cross-validation with validation and test set This is a type of k*l-fold cross-validation when l = k - 1. A single k-fold cross-validation is used with both a validation and test set. The total data set is split into k sets. One by one, a set is selected as test set. Then, one by one, one of the remaining sets is used as a validation set and the other k - 2 sets are used as training sets until all possible combinations have been evaluated. Similar to the k*l-fold cross validation, the training set is used for model fitting and the validation set is used for model evaluation for each of the hyperparameter sets. Finally, for the selected parameter set, the test set is used to evaluate the model with the best parameter set. Here, two variants are possible: either evaluating the model that was trained on the training set or evaluating a new model that was fit on the combination of the training and the validation set.

## Section 6

Applications Cross-validation can be used to compare the performances of different predictive modeling procedures. For example, suppose we are interested in optical character recognition, and we are considering using either a Support Vector Machine (SVM) or k-nearest neighbors (KNN) to predict the true character from an image of a handwritten character. Using cross-validation, we can obtain empirical estimates comparing these two methods in terms of their respective fractions of misclassified characters. In contrast, the in-sample estimate will not represent the quantity of interest (i.e. the generalization error). Cross-validation can also be used in variable selection. Suppose we are using the expression levels of 20 proteins to predict whether a cancer patient will respond to a drug. A practical goal would be to determine which subset of the 20 features should be used to produce the best predictive model. For most modeling procedures, if we compare feature subsets using the in-sample error rates, the best performance will occur when all 20 features are used. However under cross-validation, the model with the best fit will generally include only a subset of the features that are deemed truly informative. A recent development in medical statistics is its use in meta-analysis. It forms the basis of the validation statistic, Vn which is used to test the statistical validity of meta-analysis summary estimates. It has also been used in a more conventional sense in meta-analysis to estimate the likely prediction error of meta-analysis results.

## Section 7

Relative accuracy can be quantified as MSE ( λ i ) / MSE ( λ R ) {\displaystyle {\mbox{MSE}}(\lambda _{i})/{\mbox{MSE}}(\lambda _{R})} , so that the mean squared error of a candidate λ i {\displaystyle \lambda _{i}} is made relative to that of a user-specified λ R {\displaystyle \lambda _{R}} . The relative simplicity term measures the amount that λ i {\displaystyle \lambda _{i}} deviates from λ R {\displaystyle \lambda _{R}} relative to the maximum amount of deviation from λ R {\displaystyle \lambda _{R}} . Accordingly, relative simplicity can be specified as ( λ i − λ R ) 2 ( λ max − λ R ) 2 {\displaystyle {\frac {(\lambda _{i}-\lambda _{R})^{2}}{(\lambda _{\max }-\lambda _{R})^{2}}}} , where λ max {\displaystyle \lambda _{\max }} corresponds to the λ {\displaystyle \lambda } value with the highest permissible deviation from λ R {\displaystyle \lambda _{R}} . With γ ∈ [ 0 , 1 ] {\displaystyle \gamma \in [0,1]} , the user determines how high the influence of the reference parameter is relative to cross-validation. One can add relative simplicity terms for multiple configurations c = 1 , 2 , . . . , C {\displaystyle c=1,2,...,C} by specifying the loss function as

## Section 8

Example: linear regression In linear regression, there exist real response values y 1 , … , y n {\textstyle y_{1},\ldots ,y_{n}} , and n p-dimensional vector covariates x1, ..., xn. The components of the vector xi are denoted xi1, ..., xip. If least squares is used to fit a function in the form of a hyperplane ■ = a + βTx to the data (xi, yi) 1 ≤ i ≤ n, then the fit can be assessed using the mean squared error (MSE). The MSE for given estimated parameter values a and β on the training set (xi, yi) 1 ≤ i ≤ n is defined as: